Alejandro Luperon

Miami, Florida * (646) 417 - 4165
alejandro.luperon@gmail.com

Education

Thayer School of Engineering at Dartmouth College - Hanover, NH - June 2012

- Received the Bachelors of Engineering with a concentration in electrical engineering

Dartmouth College – Hanover, NH - June 2012

- Received the Bachelors of Arts in Engineering Sciences

Work Experience

Horizon3.ai New York, NY/Miami, FL (Remote) - Senior Software Engineer - August 2021 to September 2026

- Designed, wrote, maintained, and enhanced our extract, transform, and load (ETL) processing and analytics system for penetration testing data, implementing business logic and aggregations using a mix of python, SQL, and neo4j, depending on the task.
- Contributor to a data platform real-time data streaming initiative, converting/refactoring various modules in Horizon3's ETL batch processing system to a collection of microservices that consumes events from a streaming service that publishes to Kafka, creating new services & adding to new services to a Kubernetes cluster, as necessary.
- Implemented RDS database auditing (pgaudit) as well as IAM RDS database authentication and user management, with appropriate grants by engineering team, for compliance.
- Bug fixed for ETL pipelines, GraphQL APIs, and penetration testing software deployed to kubernetes. Frequently gathered stack traces, proposing fixes, deploying tactical fixes, and triaging of bugs uncovered across the entire stack, including front-end, API, ETL, and penetration testing software.
- Recurring work on SQL optimizations across GQL and ETL systems through analyzing query plans, introducing indices when necessary, and refactoring of queries using common-table expressions, subqueries, and other techniques. In some cases, optimizations resulted in up to 99.99% runtime reductions.
- Collaborator in incident response and remediation, ranging from implementing permanent or tactical fixes, gathering scope of impact, or conducting root cause analysis.
- Contributor to command-line utility, implemented in bash, extensively used by clients to autonomously run & delegate pentests and other similar operations.
- Training of other members of the engineering organization on our ETL post-pentest data processing pipeline and adjacent parts of stack through lunch & learn talks, onboarding buddy program, and collaborator to dozens of new-hire engineers over several years.
- Frequently provided direct customer interaction support calls prior to Horizon3.ai having a support team (cadence decreased with the addition of a support team).

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- Languages and technologies used include Python, SQL (postgres), neo4j, GraphQL, Docker, AWS (EC2, RDS, SQS, S3, EventBridge, Cognito User Pools, Lambda), yaml, kubernetes, helm, argo, kafka, Bash scripting, JavaScript/TypeScript, and others.

JetSweat Fitness New York, NY - Software Engineer - May 2020 to January 2021 and April 2021 - March 2022 (Contract-based)

- Integrated an initial live streaming offering for interactive and real-time live fitness classes. Worked across the entire stack over 400 hours in 2020 using Ant Media Server as the streaming engine, React for the UI on the front-end, and Sails.js on the backend.
- Partners with engineering interns and other senior engineers to knowledge share and see features through to production ready.
- Developed a WebRTC mesh network and a signaling server for a prototype of fitness classes using the RTCPeerConnection web API and designed an API which enabled support to let two peers find and connect with each other.
- Coordinating and implementing a redux integration with the main platform.
- Implemented a subdomain management workflow for white label clients.

Somewhere Good Brands New York, NY - Software Engineer & CTO - February 2020 to June 2021

- Partnered with Design and Product teams to iterate on and grow initial Somewhere Good prototype and vision.
- Planned prototyping release schedule.
- Developed and deployed initial versions of our APIs and database with a stack that consisted of Flask, SQLAlchemy, Docker, Connexion, Blueprints, AWS (EC2, Postgres RDS, Elastic Beanstalk), and prototypes with AWS Fargate and ECS
- Developed a search prototype using Elasticsearch.
- Designed data model for initial release of Somewhere Good platform.
- Interviewed and hired for iOS and backend engineer positions.
- Designed a software development lifecycle based-off already existing communication patterns and meetings across all teams (consumer, design, engineering, product, marketing) to formalize feedback loops and interactions during each stage of the lifecycle.
- Prototyped live and audio streaming features using Vonage and Ion SFU.
- Collaborated with our iOS engineer and backend engineer to implement and test releases of the platform, using UIKit and Firebase App Distribution.
- Developed a SwiftUI clickable prototype of the platform.
- Conducted research between mobile development frameworks and toolkits, such as UIKit, SwiftUI, React Native, and Flutter.
- Implemented a new front-end for Ethel’s Club Website (ethelsclub.com) and updated the back-end with a Spotify Integration.

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- Implemented the front-end and back-end for Form No Form (formnoform.com) in Python (Flask, Zappa), AWS Lambda, AWS RDS (Postgres), React.js, and other technologies under the guidance of the Head Of Creative.
- Implemented a Desktop App to manage the catalog of films, specifically reordering the playlist, uploading new features to the playlist, video processing and uploading, and editing of creator information.
- Implemented the initial version of the website for Somewhere Good (somewheregood.com). Live from September 2020 to May 2021.

Restage Experiences New York, NY - Interim CTO - March 2019 to August 2020

- Building the MVP from end-to-end. Technologies used include React, React Native, Redux, Redux-Saga, AWS (EC2, S3, RDS), Java (Dropwizard), MySQL, Firebase.
- Advising the founder on engineering needs (UX, design, technical stacks, etc.)
- Advising the founder on traction, business development, and marketing needs.
- Providing assistance to the founder by developing customer profiles, refining target audiences, conducting user interviews and usability tests, and assisting on other business matters as necessary.
- Providing guidance to other engineers on the team.

Anchor Health New York, NY - Front End Lead (Contract) - July 2019 to Dec 2019

- Worked alongside the lead platform engineer and under the guidance of the CEO to build the mobile application front-end, design platform screens, and assist on other parts of the stack as needed, using React Native, Figma, Go, AWS Lambda, and other tools.

REZI New York, NY - Senior Software Engineer - April 2016 to March 2019

- Built the flagship product from scratch as a part of the founding team using technologies including Java (Jersey, Google Guice, Java 8), MySQL, Elasticsearch, Solr, R, AWS (RDS, S3, EC2, Elasticbeanstalk), HTML, CSS, Javascript.
- Established product development processes in pair with Head of Product.
- Regularly conducted research, requirements gathering and wrote documentation.

PeerLogix New York, NY - Head Software Engineer - March 2013 - June 2014

- Built mapping system to create foreign key relationships between torrent media catalogs and media databases like IMDb and MusicBrainz.
- Built a platform that enabled aggregations of media data, by location, category, genre, and other filters.
- Tools used include Angular 1.x, Ruby on Rails, Google Maps API, Postgres, Elasticsearch, CSS, AWS, Bash, Cron

Arcadia Solutions Burlington, MA - SQL Developer & Technical Consultant - March 2013 - June 2014

- Developed SQL code used to perform analysis and data mining on Healthcare data. Specifically, developed clinical reports that quantify both EMR utilization and quality of care for payer and provider clients. Built on the Arcadia QDW framework, the reports can be executed on demand and encompass a range of clinical quality measures, including Meaningful Use, ACO, PCMH, and NQF.

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Specific responsibilities included design; build; and unit, integration, and user acceptance testing.

Partial List of Languages/Skills

- Java, Javascript, Python, and Ruby. Some proficiency with Go and Swift. Familiar with C, C++, Scala.
- Python frameworks, toolkits, and technologies, including but not limited to Flask, SQLAlchemy
- Relational databases (PostgreSQL, MySQL, etc).
- Graph databases (neo4j)
- Dependency Injection (Guice)
- React.js, Redux, Google Maps API, Charts.js, D3.js, Ajax, jQuery, HTML, and CSS
- MVC, MVVM, MVP design patterns
- Firebase, Firebase Functions
- Elasticsearch. Logstash. Familiar with MongoDB
- Amazon EC2, RDS, Aurora, S3, Elastic Beanstalk, Redshift, EventBridge, Cognito (User Pools, Identity Pools), SQS, Route 53, DynamoDB, Lambda, among other services
- Bash
- WebRTC and WebRTC technologies (Ant Media Server, Ion SFU). Knowledgeable of other P2P protocols and relevant technologies (tTorrent, bittorrent-dht, etc.)
- Argo, Kubernetes, Helm, K9s, Terraform, Crossplane